

Wind tunnel investigation of the downwash effect of a rooftop structure on plume dispersion

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ABSTRACT

This paper investigates the downwash effect of a rooftop structure (RTS) representing a typical RTS on plume dispersion. The effect of wind direction, exhaust speed, stack location, stack height, and RTS crosswind width on the severity of the downwash effect on the plume is assessed. Wind tunnel experiments were conducted to obtain plume centerline concentrations on the roof of typical low-rise and high-rise buildings. Measurements were obtained downwind of an RTS with height $h = 4$ m, along-wind length $l = 8$ m for 3 crosswind widths $w = 10$ m, 20 m and 30 m. Flow visualization was also conducted to obtain a qualitative assessment of the flow downwind of the RTS. The downwash produced by the RTS caused a significant increase in roof level concentration depending on building height, stack location, stack height, exhaust speed, wind direction and RTS crosswind width. An attempt is made to provide design guidance for determining stack height required to avoid the downwash effect for an exhaust placed downwind of the RTS.

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1. Introduction

Due to complex air flow patterns around buildings, exhaust discharged from a building roof may get re-entrained into the same building or surrounding buildings. This may lead to poor indoor air quality and potential health concerns for people working in these buildings. The presence of a rooftop structure (RTS) affects air flow patterns around the building and thus may contribute to exhaust re-ingestion problems. In particular, the downwash effect of an RTS on a plume could worsen air quality by increasing pollutant concentration near a fresh air intake. However, the effect of an RTS on plume is often overlooked in the design process, which may lead to inadequate stack height and exhaust speed.

The downwash effect of a building on plume dispersion has been widely explored [e.g. Barrett et al. (1978), Huber (1988, and 1991), Thompson (1993), Snyder and Lawson (1994), Snyder (2005)]. A review of some recent building downwash studies was provided by Canepa (2004). Downwash occurs when the plume released from a stack is brought downward by the flow of air. These studies have shown that, depending on the stack location relative

to the upstream building, ground level concentrations could be significantly higher in the presence of the building. They also showed that the downwash effect is stronger for oblique winds compared to winds perpendicular to the building leading face.

The majority of the previous research pertaining to building exhaust dispersion has been conducted for simple block-type buildings [e.g. Schulman and Scire (1991), Thompson (1993), Wilson et al. (1998)]. These studies have shown that an increase in stack height or exhaust speed decreases roof-level concentrations. A few studies have assessed the effect of various types of RTS on building exhaust dispersion (e.g. Stathopoulos et al., 2002, 2003, 2004), Petersen et al. (1999) and Lowery and Jacko (1996), the focus of these studies was not on the downwash effect of the structure. Britter and Hanna (2003) provided a detailed overview of flow and dispersion in urban areas.

Studies on groups of buildings cover a wide range of building arrangements, from case specific studies of industrial or urban groups of buildings, to investigations of idealized regular arrays of obstacles, as reviewed in detail by Hall et al. (1996). Macdonald et al. (1997) and Hall et al. (1996) measured the mean velocity profiles within a regular cubical array using pulsed wire anemometry, and found a rapid reduction in speed with increasing area density. The influence of the buildings (with height H) was found to extend up to $2-3H$, which is to be expected since the effect of a single building generally extends up to about $3H$. Macdonald (1997 and 1998) compared the relative spreads of plumes in open terrain and within arrays. He found that lateral and vertical plume spreads σ_y and σ_z

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were enhanced within the array due to the increased levels of turbulence produced by the buildings. A variety of semi-empirical methods for calculating σ_y and σ_z as a function of z_0 using various Gaussian models have been reviewed in Hall et al. (1996).

Few studies have looked into the downwash effect of an RTS on plume dispersion [for e.g. Wilson (1979), Saathoff et al. (2002, 2003a,b and 2009) and Gupta et al. (2005a, 2005b and 2006)]. Wilson (1979) showed the formation of recirculation zones on buildings with various RTSs. However, no concentration measurements were presented to quantify the effect of the RTS on plume dispersion. The influence of the downwash effect of an RTS was investigated by Saathoff et al. (2002) using full-scale tests. Concentration measurements were obtained on the roof of a typical low-rise building from a stack located downwind of an RTS. Results showed that plume rise decreased drastically causing a significant increase in roof level concentrations (i.e. drop in dilutions). In general, roof-level concentrations increased by up to 10 times due to the downwash effect of the RTS. However, the findings were limited to one stack height and one exhaust speed. Gupta et al. (2005a,b and 2006) showed similar results using wind tunnel tests.

Even though the basic downwash behavior due to RTS is similar to that for a building, the effect on dispersion is not necessarily the same. Downwash causes increase in roof level concentrations in the RTS case and also ground level concentration in the building case but there are also differences. For instance, in the RTS case, the present study found that increase in RTS width increases rooftop concentrations; however Huber (1988) showed that in the case of building downwash ground level concentrations decrease when increasing the building width. Clearly, the building downwash knowledge base is not necessarily applicable on RTS downwash.

The present study aims at quantifying the downwash effect of an RTS on plume dispersion using wind tunnel experiments. Concentration measurements obtained in the wind tunnel for different building configurations are presented. The effects of RTS crosswind width, building height, stack height, exhaust speed, wind direction and separation distance between stack and RTS on the severity of downwash are discussed.

2. Wind tunnel modeling criteria

Various criteria are required for modeling dispersion in a wind tunnel. These requirements are provided in the ASCE (1999) – Manual of Practice No. 67 on “Wind Tunnel Studies of Buildings and Other Structures”. The following criteria are generally recommended for non-buoyant exhausts in wind tunnel dispersion studies:

- similarity of the wind tunnel boundary layer with the atmospheric surface layer (i.e. similar mean velocity and turbulence intensity profiles);
- building Reynolds number (Re_b) should be greater than 11,000 for sharp edged structure, where $Re_b = U_H l / \nu$;
- stack Reynolds number (Re_s), should be greater than 2,000, where $Re_s = w_e d_e / \nu$;

where U_H is the wind speed at building height, l is the smaller of building height and cross-wind width, and ν is the kinematic viscosity of air, d_e is the stack inside diameter, and w_e is the exhaust speed.

The building Reynolds number criterion ensures that the flow patterns around the model and full-scale building are similar. The stack Reynolds number ensures that the exhaust discharged from the stack is turbulent. Further discussion of wind tunnel modeling criteria is provided in fluid modeling guidelines by Isyumov and Tanaka (1979) and Snyder (1981).

3. Experimental methodology

Wind tunnel experiments were carried out at Concordia University's boundary layer wind tunnel. The wind tunnel working section is 1.8 m by 1.8 m and the length is 12.2 m. Complete details on the wind tunnel may be found in Stathopoulos (1984). The building models were constructed at a scale of 1:200 and had a square plan with a full-scale side dimension of 50 m. The full-scale building heights (H) were 15 m representing a typical low-rise building, and 60 m representing a typical high-rise building. A brief survey of laboratory and hospital buildings was conducted to determine a typical size of RTS. Based on the results of this survey an RTS 4 m in height (h) and 8 m along-wind length (l) with 3 crosswind widths – 10 m, 20 m, and 30 m was used for testing.

A brass stack with an equivalent full-scale diameter (d_e) of 0.6 m was located on the roof of the building at a distance of $0.4L$ from the leading edge of the building where L is the width of the building. The stack height (h_s) was variable and had values of $0.25h$, $0.75h$, $1.25h$, and $1.75h$, which correspond to full-scale heights of 1 m, 3 m, 5 m, and 7 m, respectively. The separation distance (x_s) between the stack and RTS varied from $0.5h$ to $2.5h$. Experiments were carried out for wind directions ranging from 0° to 60° . However, the majority of tests were conducted for $\theta = 0^\circ$ and 45° . Fig. 1 shows the positions of the stack, rooftop structure and the sampling locations.

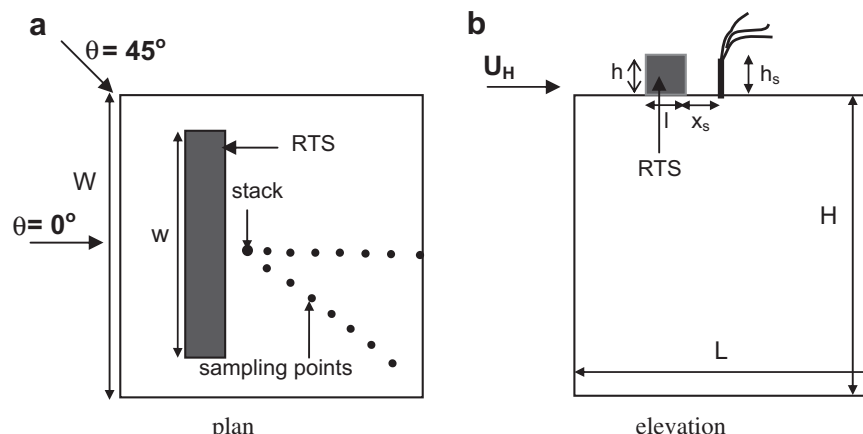


Fig. 1. Schematic view of building, RTS, and sampling locations.

3.1. Boundary layer simulation

An urban boundary layer was simulated for the present study. Vertical profiles of mean velocity and longitudinal turbulence intensity were obtained with a TSI hot-wire anemometer at a model scale of 1:200 and are presented in Fig. 2a and b, respectively.

A model widely used to describe the mean velocity profile in the atmospheric boundary layer is the power law.

$$\frac{U(Z)}{U(Z_g)} = \left(\frac{Z}{Z_g}\right)^\alpha \quad (1)$$

where α is the power law exponent, $U(Z)$ is the wind speed at any height Z and $U(Z_g)$ is the wind speed at gradient height (Z_g). The mean velocity profile for the urban exposure had a power law exponent (α) of 0.31. The estimated model-scale roughness length (z_o) was approximately 5.5 mm. At a model scale of 1:200, the equivalent full-scale roughness (z_o) is 1.1 m, which is within the expected range for an urban environment ($0.5 \text{ m} < z_o < 1.5 \text{ m}$) recommended by Wieringa (1993). Another consistency check is to relate α to z_o . Counihan (1975) showed that α and z_o are related by the following expression:

$$\alpha = 0.24 + 0.96 \log_{10} z_o + 0.016 (\log_{10} z_o)^2 \quad (2)$$

For $z_o = 1.1 \text{ m}$, Eq. (2) gives an α value of 0.25, which is approximately 20% lower than the measured value of 0.31.

The longitudinal turbulence intensity $I_u(Z)$ at height Z is defined with the following expression.

$$I_u(Z) = \sigma_u(Z)/U(Z) \quad (3)$$

where σ_u is the standard deviation of longitudinal component of wind speed. The I_u values observed in the upwind urban boundary layer at the model building heights were 27% for the low-rise building ($H = 15 \text{ m}$) and 18% for the high-rise building ($H = 60 \text{ m}$), as shown in Fig. 2.

3.2. Flow visualization

Flow visualization was carried out prior to concentration measurements to qualitatively assess the downwash effect of an RTS on plume dispersion. A Dantec fog generator was used to

produce smoke using a water-based fluid specially formulated for exhaust dispersion applications. Smoke was released from a stack with $h_s = 1.25h$ (5 m) located 2 m downwind of the RTS with $w = 30 \text{ m}$ and $h = 4 \text{ m}$. The photographs were taken for different configurations with a digital camera. The visualization tests were conducted for a constant upstream wind speed of 5 m s^{-1} at the building height. The exhaust speed was not measured. However, it was kept low to ensure that the plume rise was negligible.

3.3. Exhaust simulation

The present study deals only with non-buoyant exhausts. Sulfur hexafluoride (SF_6) was used as the main tracer gas. Although SF_6 is heavier than air, to simulate a non-buoyant exhaust, a certified mixture of sulfur hexafluoride (SF_6) and nitrogen was emitted from an adjustable brass stack. The concentration of tracer gas was very low – 10 ppm. Thus, buoyancy effects were negligible. Along-wind concentration measurements were obtained at seven sampling points located at distances of $0.62h$, $1.25h$, $2.50h$, $3.75h$, $5h$, $6.25h$ and $7.0h$ from the stack. For both normal and oblique winds, the sampling locations were in line with the wind direction. The spanwise and vertical concentration profiles for the low-rise building were obtained at a distance of $2h$ and $5h$ from the stack. For the crosswind profile, the concentration measurements were obtained at seven sampling locations separated by a distance of $1.25h$. The vertical dilution profiles were obtained by using a thin vertical brass tube with sampling tubes attached at different heights. Concentration measurements were obtained for z values ranging from $z = 0$ (roof level) to $3h$.

A syringe sampler was used to obtain samples of 1 min duration, which typically represents an averaging time of 15–30 min at full-scale [Wilson and Britter (1982)]. A key parameter for simulating plume dispersion at a model scale is the exhaust momentum ratio, $M[(\rho_e/\rho_a)^{0.5}(w_e/U_H)]$, i.e. the ratio of the square root of the momentum fluxes, where ρ_e is the density of the exhaust and ρ_a is the ambient air density. Since the measurements were carried out with non-buoyant exhausts, $\rho_e = \rho_a$ and $M = w_e/U_H$. The M values ranged from 1 to 5, representing strong to moderately weak wind speeds. The exhaust speed ranged from a minimum of 5.4 m s^{-1} ($M = 1$) to a maximum 27 m s^{-1} ($M = 5$). The corresponding range of stack Reynolds number is 1100–5500. The criterion of $\text{Re}_s > 2000$

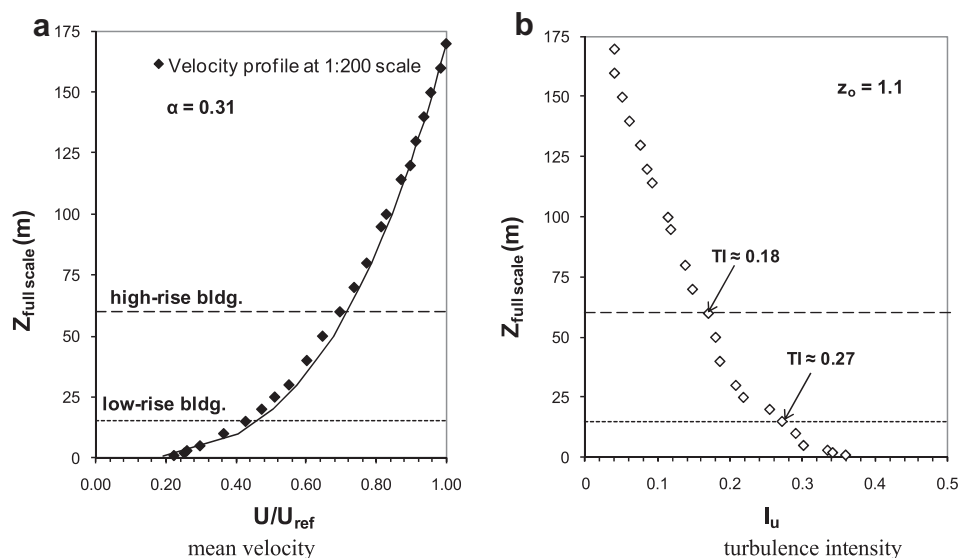


Fig. 2. Mean velocity and longitudinal turbulence intensity profiles.

for stack Reynolds number independence may not be met for $M = 1$, indicating that the flow was laminar for this case. Wilson et al. (1998) suggested a correction factor, which could be used to determine the full-scale equivalent M corresponding to the model value. He suggested that for laminar flows from stacks:

$$M_{\text{full-scale}} = 1.414M_{\text{model}} \quad (4)$$

Therefore, according to Eq. (4), $M_{\text{model}} = 1$ in the present study is equivalent to $M_{\text{full-scale}}$ of 1.4. Such small variation in M value is not expected to produce significant discrepancies. Therefore, no attempt was made to modify the exhaust to increase the Re_s . In the present study the repeatability of concentration measurements generally ranged between $\pm 10\%$ and $\pm 20\%$.

3.4. Normalized dilution

The concentration data obtained from wind tunnel experiments were converted to minimum dilution values using the following formula:

$$D_{\min} = C_e/C \quad (5)$$

where C is the receptor concentration and C_e is the exhaust concentration.

Note that the term “dilution”, as defined in Equation (5), is not consistent with the definition in standard atmospheric dispersion textbooks such as Pasquill (1974), where “dilution” refers to the effect of the mean wind speed on diluting the plume and its centerline concentration. With this definition, a reduction in dilution” could be caused by downwash and have nothing to do with wind speed and actual dilution. Thus to take into account the effect of mean wind speed, the dilutions were converted to normalized dilution D_N , as suggested by Wilson et al. (1998). The dilutions were normalized by the exhaust flow rate (V), the wind speed at the building height, and the square of the building height. The expression for the normalized dilution is:

$$D_N = \frac{D_{\min} V}{U_H H^2} \quad (6)$$

The usefulness of D_N derives from the fact that it remains relatively constant for a variety of full-scale atmospheric conditions and varying ratio of exhaust velocity to wind speed. In the present study, the wind speed at the building height was kept constant and exhaust speed was varied to simulate different values of momentum ratio. Increasing the exhaust speed increases the flow rate as well. Thus the results presented in this paper are in the form

of normalized dilution which takes into account the effect of wind speed.

It should be noted that plume rise, h_r , for a momentum jet, can be estimated using the following formula developed by Briggs (1984): $h_r = 3Md_e$. Since h_r depends only on the exhaust momentum ratio, the results of the current study are applicable for different wind speeds.

4. Results and discussion

Flow visualization photos obtained with the test buildings with and without an RTS are presented in Fig. 3. The RTS height, h , was 4 m and its width, w , was 30 m ($7.5h$). The experiments were carried out with a 5 m tall stack ($h_s = 1.25h$) located 2 m downwind of the RTS ($x_s = 0.5h$).

The smoke patterns obtained with the low-rise building for $\theta = 0^\circ$ and $\theta = 45^\circ$ are shown in Fig. 3a. Note that for the flat-roofed building, the plume did not make significant contact with the roof for either wind direction. However, for the building with the RTS, the plume was deflected downwards and a significant portion of it made contact with the roof. The downwash effect of the RTS on the plume is clearly evident.

Flow visualization photographs for the tall building with and without the RTS are shown in Fig. 3b for $\theta = 0^\circ$. Note that, due to increase in height a large separation bubble is formed on the roof of tall building compared to the low-rise building. In this case, the plume and RTS were completely engulfed by the bubble. Thus, an RTS totally engulfed inside the bubble has little effect on dispersion of exhaust from a rooftop source.

4.1. Effect of RTS crosswind width on along-wind D_N profiles

The effect of RTS crosswind width on along-wind D_N profiles is shown in Fig. 4–6. Results are presented for a 3 m ($0.75h$) stack with $M = 2$ located 2 m ($0.5h$) downwind of the RTS. Measurements are shown for 3 crosswind widths, $w = 10$ m ($2.5h$), 20 m ($5h$) and 30 m ($7.5h$) for the low-rise and high-rise building. Data for the flat-roofed buildings are also shown for comparison. The distance between the sampler and the stack, x , is normalized with the height of the RTS.

4.1.1. $\theta = 0^\circ$

The D_N profiles for the low-rise building for $\theta = 0^\circ$ are shown in Fig. 4a. For this case, the addition of an RTS increased concentrations at the roof level (reduced D_N values) significantly, regardless of its width, especially for small values of x , $x < 15$ m. Even with the

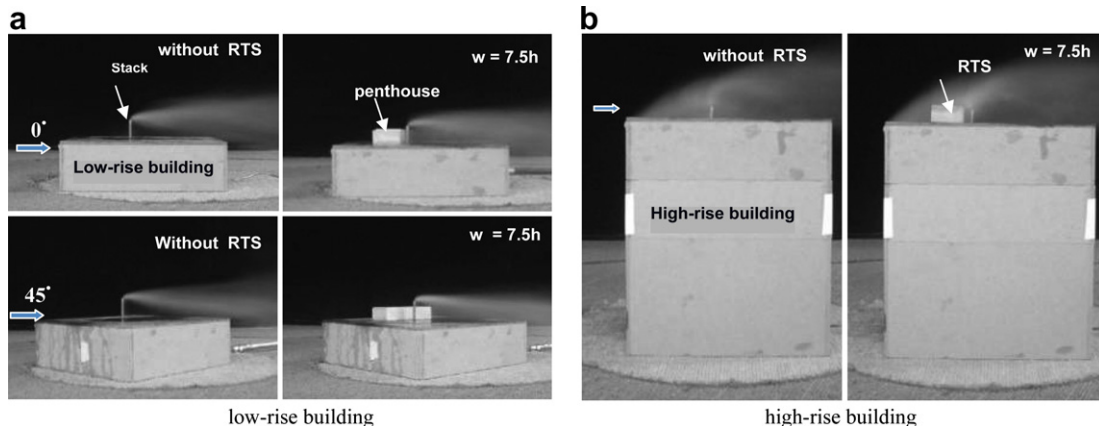


Fig. 3. Flow visualization showing the effect of RTS on plume.

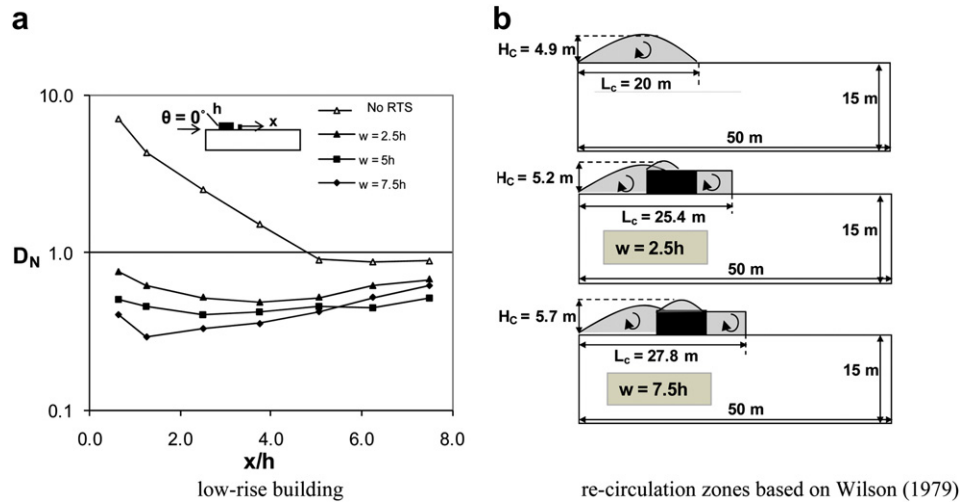


Fig. 4. Effect of RTS cross-wind width on along-wind D_N profile for the low-rise building: $\theta = 0^\circ$, $h_s = 0.75h$, and $M = 2$.

smallest width ($w = 2.5h$), concentrations increased by a factor of 10 at the sampler located closest to the stack. Dilutions reduced with increase in RTS width with a maximum reduction of approximately 20 times with $w = 7.5h$. The effect of RTS width on dilutions was generally similar for other h_s and M values. However, the severity of the downwash varied with stack height and exhaust momentum ratio, as further discussed in subsequent sections.

Roof level dilutions decreased with the RTS due to entrainment of the plume within the RTS downwind recirculation zone. Fig. 4b shows the dimensions of recirculation zones for the low-rise building and RTS, as estimated from equations suggested by Wilson (1979). The zones are shown for the narrow ($w = 2.5h$) and wide RTS ($w = 7.5h$). In brief, Wilson's model predicts the height and length of the building roof recirculation zone (separation bubble) as $H_c = 0.22R$, and $L_c = 0.9R$. The parameter R is known as the scaling length and is given by, $R = B_s^{0.67} B_L^{0.33}$, where B_s is the smaller of the building upwind face dimensions (height or width) and B_L is the larger dimension. For the low-rise building $B_s = 15$ m and $B_L = 50$ m, producing $R = 22.3$ m. Note that H_c increased to 5.7 m from 4.9 m and L_c increased to 27.8 m from 20 m in presence of the wide RTS ($w = 7.5h$). A larger downwind cavity will entrain more of the plume causing a higher reduction in roof level dilutions with the wide RTS.

Normalized dilution profiles obtained for the tall building are shown in Fig. 5a. The dilution values with and without the RTS are nearly the same, indicating virtually no influence of the RTS on the plume for the tall building. The minor differences in concentration values with different RTS crosswind widths can be attributed to the repeatability error, which were between $\pm 10\%$ and $\pm 20\%$ in the present study. The dimensions of the separation bubble for the tall building are shown in Fig. 5b where $B_s = 50$ m and $B_L = 60$ m. The building length scale (R) is 53.1 m, giving bubble dimensions $H_c = 11.7$ m and $L_c = 47.8$ m, which are more than twice as large as the values for the low-rise building. In this case, the RTS and plume are engulfed within the recirculation zone. Consequently, the RTS did not significantly influence the upstream flow, as indicated by flow visualizations shown in Fig. 3b.

4.1.2. $\theta = 45^\circ$

Fig. 6a shows the D_N profiles for the low-rise building for the oblique wind ($\theta = 45^\circ$). For this case, D_N values for the flat-roofed building are nearly 10 times higher than dilutions obtained for $\theta = 0^\circ$. Previous studies such as Wilson and Winkel (1982) and Schulman and Scire (1991) also reported an increase in roof level dilutions for $\theta = 45^\circ$. This happens due to the absence of the separation bubble that forms at the roof leading edge for $0^\circ \leq \theta < \pm 30^\circ$.

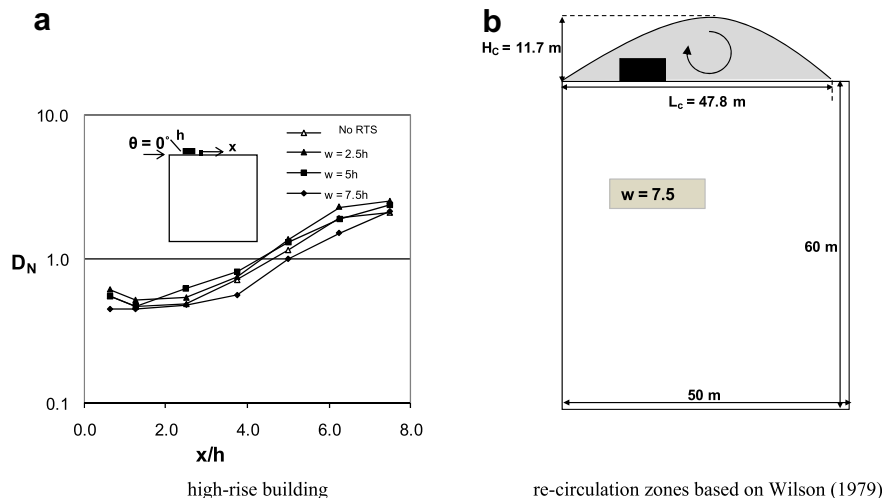


Fig. 5. Effect of RTS cross-wind width on along-wind D_N profile for the high-rise building: $\theta = 0^\circ$, $h_s = 0.75h$, and $M = 2$.

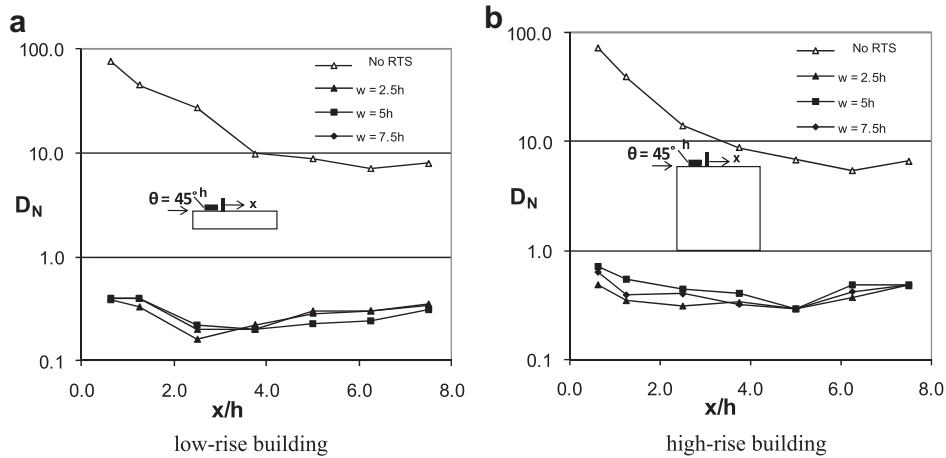


Fig. 6. Effect of RTS cross-wind width on along-wind D_N profiles for $\theta = 45^\circ$; $h_s = 0.75h$ and $M = 2$.

For $\theta > 30^\circ$, delta wing vortices are formed around the building upwind edges. Thus, for an oblique wind, a plume emitted in the central region of a roof without an RTS will not be affected significantly by building-generated turbulence. Consequently, vertical plume spread is diminished, yielding relatively high dilutions at the roof surface.

D_N values for the low-rise building with RTS are significantly lower than those obtained with the flat-roofed building. Near the stack, D_N is reduced by a factor of 200; near the leeward edge of the building, about the reduction factor is approximately 25. It should be noted that D_N shows little dependence on distance from the stack or RTS width. In general, for short stacks ($h_s \leq 0.75h$), D_N values for $\theta = 45^\circ$ were approximately half as large as those obtained for $\theta = 0^\circ$, indicating a stronger downwash effect for the oblique wind direction. However, for tall stacks ($h_s \geq 1.25h$), D_N values were similar for both wind directions.

Fig. 6b shows the dilution profiles obtained for the high-rise building for $\theta = 45^\circ$. The dilution profiles are generally similar to those obtained for the low-rise building for $\theta = 45^\circ$. The D_N values obtained with the RTS are significantly lower than those obtained with the flat-roofed building and show little dependence on x or w .

4.2. Effect of RTS on crosswind plume spread

Span-wise concentration measurements were obtained for the low-rise building to demonstrate the effect of the RTS on crosswind plume spread. In order to compare the results with the classical Gaussian model, the results are expressed as normalized concentration (C_N), which is defined as:

$$C_N = CU_H H^2 / (C_e Q) = 1/D_N \quad (7)$$

The lateral plume spread (σ_y) values were estimated assuming that the lateral concentration profile follows a Gaussian distribution. The expression used to fit a Gaussian distribution is:

$$C(y) = C_{\max} \left\{ \exp \left[-\frac{(y + y_c)^2}{2\sigma_y^2} \right] \right\} \quad (8)$$

where $C(y)$ is the spanwise concentration at distance y from the plume centerline, $C_{\max}$ is the maximum concentration and y_c is the lateral displacement of the plume about the approaching wind direction.

Fig. 7a and b show the lateral C_N profiles obtained for $\theta = 0^\circ$ at $x = 2h$ and $x = 5h$, respectively. The results are shown for $M = 2$ and

$h_s = 0.75h$. Note that the profiles are symmetrical about the approaching wind direction. Therefore, $y_c = 0$ for this case. At $x = 2h$, the σ_y value for the building with the RTS increased by more than 50% compared to that for the flat-roofed building. This is due to entrainment of the plume into the near wake of the structure. The effect of the RTS on the plume is less significant farther from the stack. Therefore, at $x = 5h$, the σ_y value increased only marginally with the RTS, as shown in Fig. 7b.

The lateral C_N profiles for $\theta = 45^\circ$ are shown in Fig. 8. For this case, the σ_y values obtained at both $x = 2h$ and $x = 5h$ without the RTS were about 10%–15% smaller than those for $\theta = 0^\circ$. This suggests that for a building where the plume is not influenced by an upwind RTS, the plume is narrower for the oblique wind than that for the normal wind. With the RTS, the σ_y value increased by 70% at $x = 2h$, and by nearly 50% at $x = 5h$. It is interesting to note that with the RTS, $C_{\max}$ was located approximately 5 m off the center-line of the roof. Therefore, in order to obtain a better fit with the Gaussian model, $y_c = -5$ m.

Fig. 9 taken from Snyder (2005) shows the flow fields around a building with $W = 4H$ for normal and oblique wind directions. Note that the flow downwind of the building is symmetrical for the normal wind and asymmetrical for the oblique wind. Similar or even more complex flow patterns are expected to occur on the roof of a building downwind of the RTS. Thus, a lateral shift in mean plume trajectory with the RTS for an oblique wind is not surprising.

Some dispersion models (e.g. ADMS and AERMOD) include approximations (based on wind tunnel observations) relating vertical and lateral plume spreads with turbulent spread and building dimensions. The UK Urban Dispersion Model (UDM) contains simple parameterizations based on wind tunnel studies. A variety of semi-empirical methods for calculating σ_y and σ_z as a function of z_0 using various Gaussian models have been reviewed in Hall et al. (1996). However, it has been found out that such models are not generally suitable for rooftop exhausts or in general micro-scale dispersion (i.e. distances less than 100 m), let alone rooftop structures or other architectural configurations and details, Hajra et al. (2010). Modifying these models to make them usable for building roof exhausts would be a separate (and rather ambitious) undertaking. The models that are recommended for rooftop dispersion are the dilution models given by ASHRAE (2011).

4.3. Effect of RTS on plume height

For the low-rise building, vertical dilution profiles were obtained to determine the effect of the RTS on plume height (h_p),

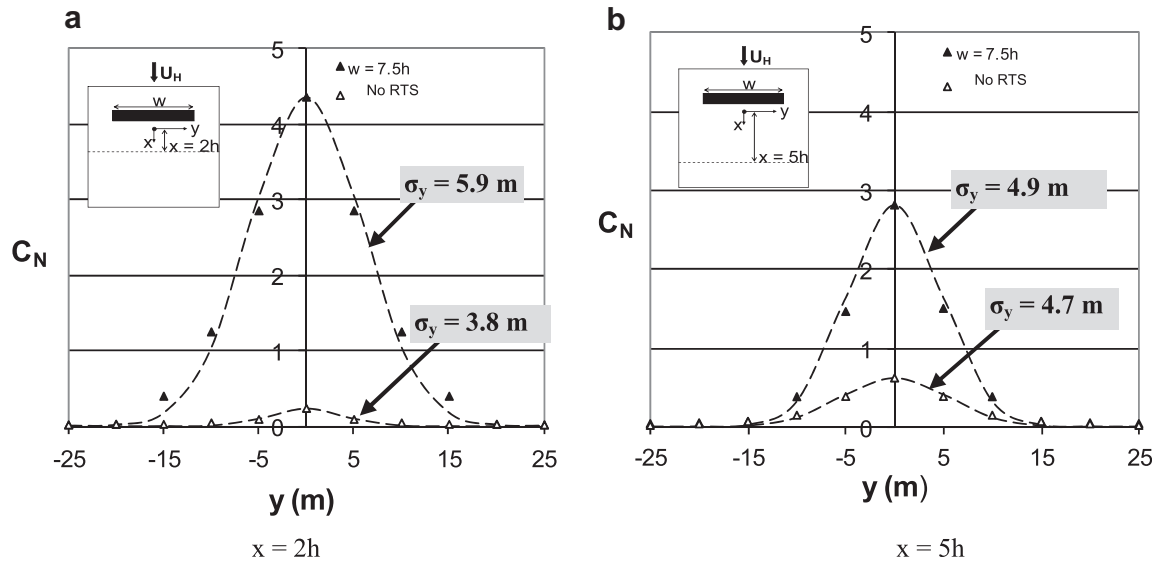


Fig. 7. Crosswind profiles of normalized concentrations (C_N) for the low-rise building with and without the RTS ($w = 7.5h$) for $\theta = 0^\circ$, $h_s = 0.75h$ and $M = 2$.

which is the height at which the minimum D_N (i.e. maximum plume centerline concentration) occurs. The vertical dilution profiles obtained at $x = 2h$ and $x = 5h$ for $\theta = 0^\circ$ are shown in Fig. 10a and b, respectively. The results are shown for $x_s = 0.5h$, $M = 2$ and $h_s = 0.75h$.

A decrease in plume height at both locations for the building with the RTS is evident. The values of h_p for the building with and without the RTS were $0.9h$ and $1.2h$, respectively. Thus, for this stack location, the presence of the RTS reduced h_p by 30% for $\theta = 0^\circ$. The vertical dilution profiles for $\theta = 45^\circ$ are shown in Fig. 11a and b for $x = 2h$ and $x = 5h$, respectively. In this case h_p is $0.7h$ with the RTS, a reduction of 40% compared to the flat roof case. The greater reduction in plume height for $\theta = 45^\circ$ indicates that the downwash effect of the RTS is greater than that for $\theta = 0^\circ$.

Note that the vertical D_N profiles for the flat-roofed building are relatively symmetrical about the plume height. However, for the building with the RTS, the D_N profiles are not symmetrical. This is due to plume reflection, as it makes contact with the roof [Turner (1994)]. Because of the reflection of the plume at the roof surface, the samplers located close to the RTS experience significantly higher concentrations than those on the flat roofed building, which has an elevated plume.

4.4. Effect of stack height on normalized dilutions

The effect of stack height on normalized dilutions for the low-rise and high-rise building with and without the RTS is shown in Fig. 12–15. Results are shown for $h_s = 0.25h$, $0.75h$, $1.25h$ and $1.75h$ for a low M value ($M = 2$) and a high M value ($M = 5$) for $\theta = 0^\circ$ and $\theta = 45^\circ$. Measurement data are shown for two sampling locations, $x = 1.3h$ and $x = 7h$ for $w = 2.5h$ and $w = 7.5h$.

4.4.1. Low-rise building: $\theta = 0^\circ$

Measurements obtained with the low-rise building for $\theta = 0^\circ$ are shown in Fig. 12 for $M = 2$ and $M = 5$. In general, irrespective of the presence of an RTS, for both M values, an increase in stack height decreases concentration values (increases D_N values) at all sampling locations. However, the effect of h_s was more pronounced at the sampler close to the stack and for the low M value ($M = 2$).

The D_N values increased significantly for the building without the RTS. As shown in Fig. 12a, for $M = 2$, an increase in h_s from $0.25h$ to $1.75h$ increases D_N values by a factor of nearly 1000 at $x = 1.3h$ and by a factor of approximately 10 at $x = 7h$. However, for $M = 5$, D_N increases by a factor of 100 at $x = 1.3h$ and by a factor of approximately 10 at $x = 7h$. Closer to the stack, building turbulence

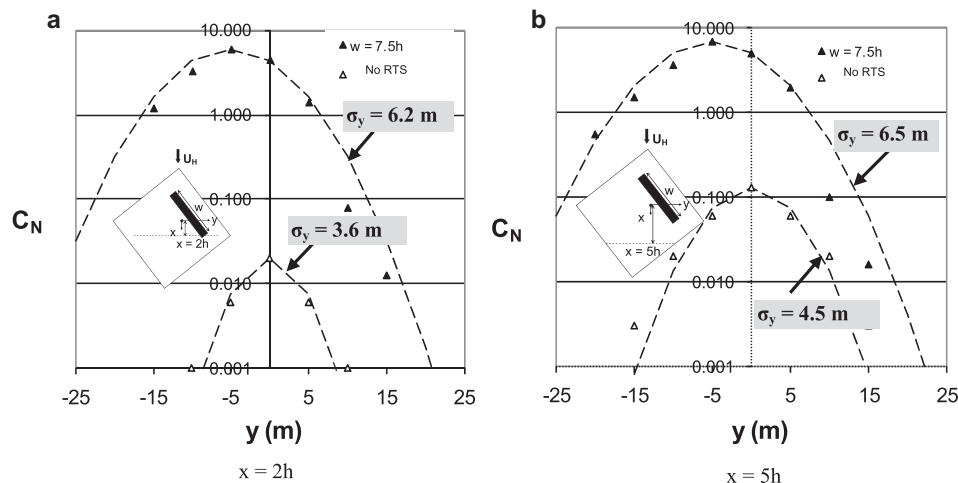


Fig. 8. Crosswind profiles of normalized concentrations (C_N) for the low-rise building with and without the RTS ($w = 7.5h$) for $\theta = 45^\circ$, $h_s = 0.75h$ and $M = 2$.

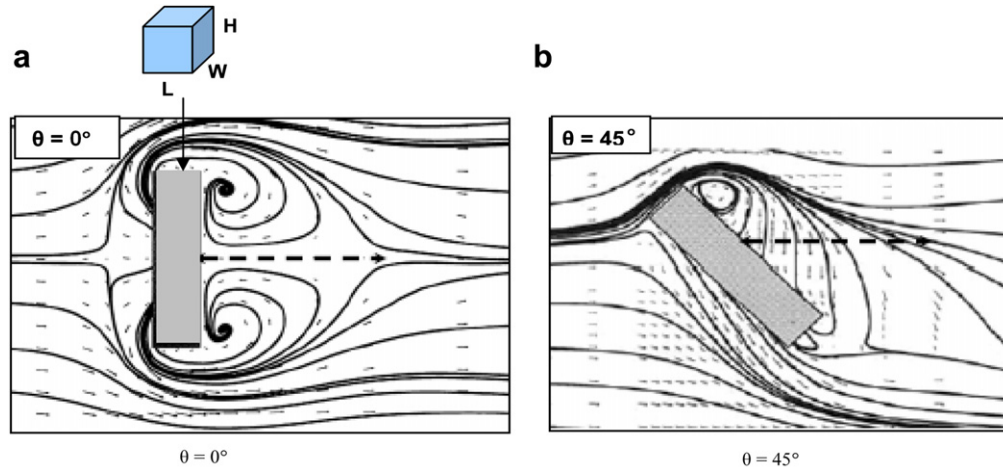


Fig. 9. Ground plane streamlines around building with $W = 4H$ for normal and oblique wind after Snyder (2005).

effects on the plume are higher than that near the leeward edge of the building. The plume spread increases with an increase in distance from the stack. Thus, the effect of h_s on D_N values is less at samplers located farther from the stack.

As with the flat-roofed building, D_N values for the building with the RTS also increases with an increase in h_s . Near the stack, the effect of h_s is greater for the narrow RTS ($w = 2.5h$) than for the wide RTS ($w = 7.5h$). For both M values, $M = 2$ and $M = 5$, the D_N values increase by a factor of about 1000 for $w = 2.5h$ compared to nearly 20 times for $w = 7.5h$ when h_s is increased from $0.25h$ to $1.75h$. At $x = 7h$, this increase in h_s caused D_N to increase by only a factor of 10.

4.4.2. Low-rise building: $\theta = 45^\circ$

The effect of h_s on D_N for the low-rise building for $\theta = 45^\circ$ is shown in Fig. 13. For both $M = 2$ and $M = 5$, concentrations decreased (increased D_N) significantly with increasing stack height. For $M = 2$, a stack height increase from $0.25h$ to $1.75h$ increases D_N by a factor of nearly 400 at $x = 1.3h$, as shown in Fig. 13a. For $M = 5$, D_N values increase by a factor of 16, as shown in Fig. 13b. The effect of stack height diminishes with increasing distance from the stack. At $x = 7h$, D_N values increase by nearly 10 times for both M values, which is similar to results obtained for the normal wind case. As

with the flat-roofed building, an increase in stack height increases D_N values for the building with the RTS. Since the effect of RTS width for $\theta = 45^\circ$ is small, D_N values for any stack height are generally similar for different RTS widths.

4.4.3. High-rise building

The effect of h_s on D_N for the high-rise building for $\theta = 0^\circ$ was nominal. A slight increase in D_N values is observed at $x = 1.3h$ while the effect is minimal at $x = 7h$. At $x = 1.3h$, D_N values increase by a factor of 6 when h_s increases from $0.25h$ to $1.75h$. Since the effect of the 4 m tall RTS is negligible for the high rise building for $\theta = 0^\circ$, the concentration data for any h_s with different RTS widths are similar to the data without the RTS. For this case, the effect of h_s on D_N was small since the plume was entrained within the rooftop recirculation zone for all h_s tested. Although a modest increase in dilution values occurred with increase in M , the effect of h_s on D_N values was still not that significant. The effect of h_s on D_N for $\theta = 45^\circ$ are generally similar to those obtained for the low-rise building for $\theta = 45^\circ$.

4.5. Effect of separation distance on normalized dilutions

The effect of RTS-stack separation distance (x_s) for the low-rise building for $\theta = 0^\circ$ and $\theta = 45^\circ$ is shown in Fig. 14a and b,

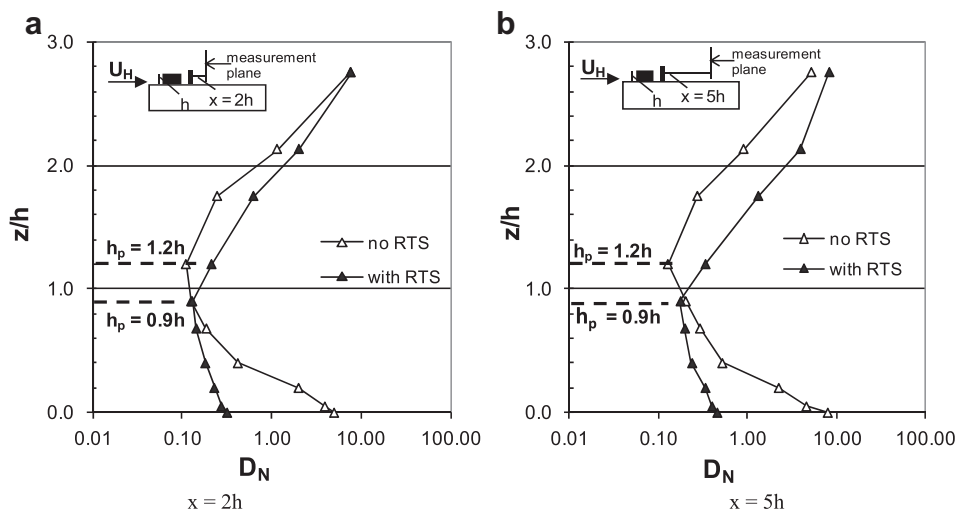


Fig. 10. Vertical profiles of normalized dilutions (D_N) for the low-rise building with and without the RTS ($w = 7.5h$) for $\theta = 0^\circ$, $h_s = 0.75h$ and $M = 2$.

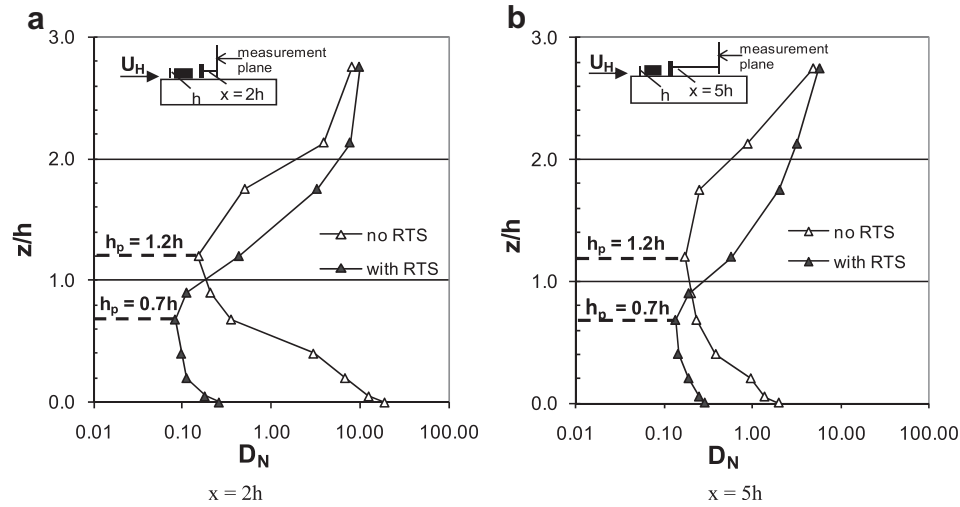


Fig. 11. Vertical profiles of normalized dilutions (D_N) for the low-rise building with and without the RTS ($w = 7.5h$) for $\theta = 45^\circ$, $h_s = 0.75h$ and $M = 2$.

respectively. Results are shown for the wide RTS ($w = 7.5h$) for $h_s = 0.75h$ and $M = 2$. The value of x_s varied from $0.5h$ to $1.5h$. It should be noted that similar trends were observed with the smaller RTS widths.

For $\theta = 0^\circ$, increasing x_s from $0.5h$ to $1.5h$ increases D_N values by approximately 4 times near the stack ($x < 2.5h$), where the downwash effect is more evident. Further from the stack, the increase in x_s produced only a modest increase in dilution. The

effect of separation distance on dilutions is greater for the oblique wind. For $\theta = 45^\circ$, increasing x_s from $0.5h$ to $1.5h$ caused D_N values to increase by approximately a factor of 10 within $x < 2.5h$. Downwash weakens farther from the RTS and thus dilutions increase with increasing x_s . The differences between $\theta = 0^\circ$ and $\theta = 45^\circ$ when the RTS is present may be due to the non-symmetrical nature of the flow around the structure for an oblique wind. Sampling locations were aligned with the approaching

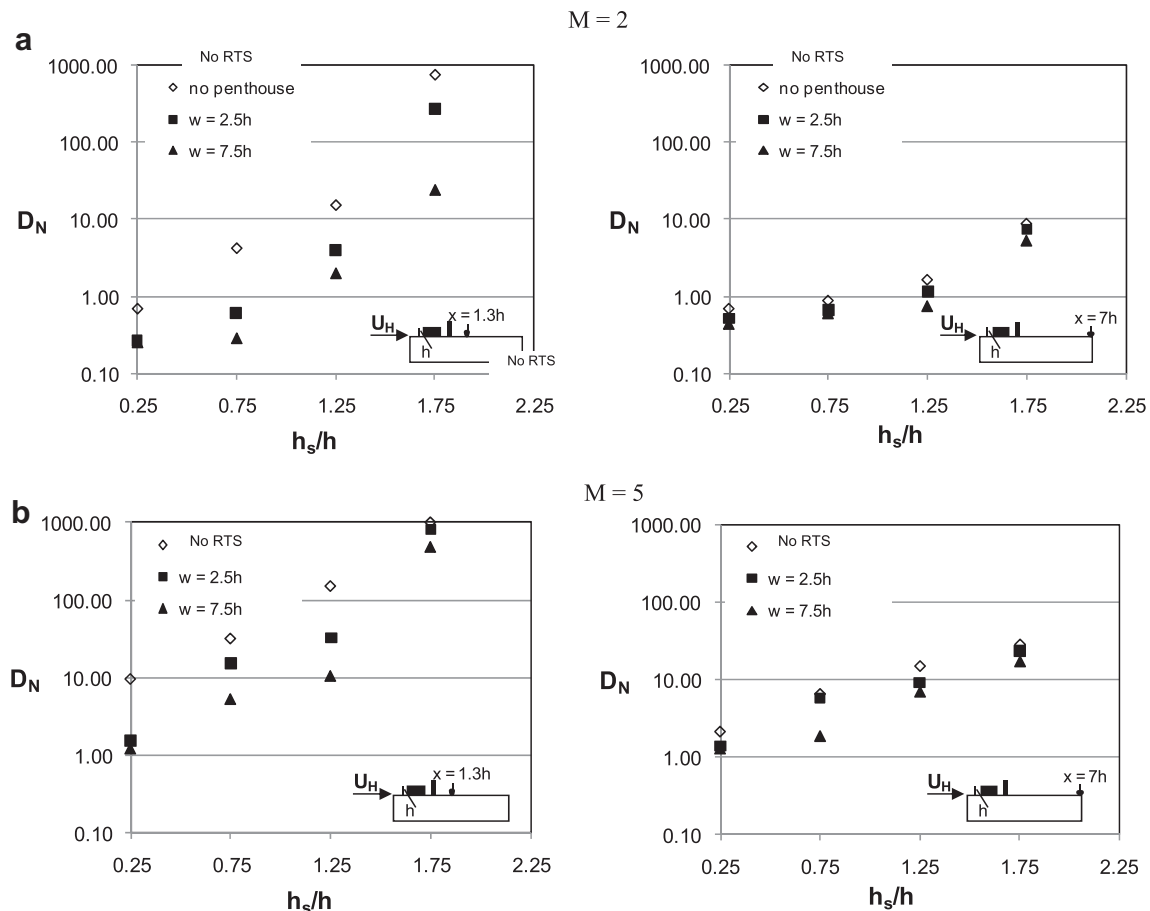


Fig. 12. Effect of stack height on D_N values for the low-rise building with and without the RTS for $\theta = 0^\circ$.

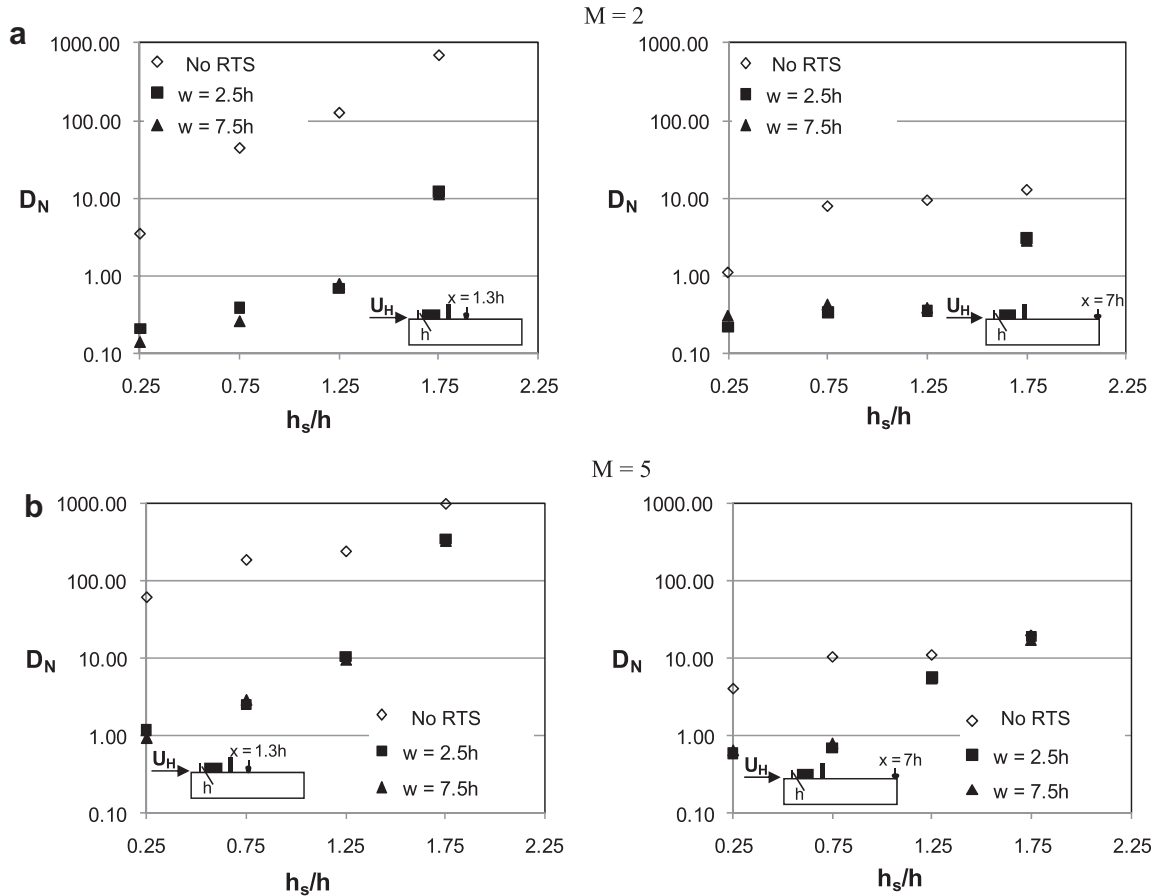


Fig. 13. Effect of stack height on D_N values for the low-rise building with and without the RTS for $\theta = 45^\circ$.

for wind direction, but for $\theta = 45^\circ$, the plume centerline may have deviated from the along-wind direction. Consequently, the uncertainty of the data for $\theta = 45^\circ$ is greater than that for $\theta = 0^\circ$.

4.6. Stack height required to avoid the downwash effect of an RTS on plume

Previous studies have provided design aids or rules of thumb to estimate the required stack height in order to avoid the downwash

effect of a building on a downwind stack. Lucas (1972) suggested that the stack must be 2.5 times the height of the nearest tall building. Snyder and Lawson (1976) showed that Lucas' rule is justified for wide buildings ($W > 2.5H$). However, for slender structures ($W \leq 2.5H$), Lucas' rule could be relaxed to $(H + 1.5l)$, where l is the smaller dimension of building width or height. Petersen et al. (1999) suggested a stack height reduction factor (SHR) to estimate the required stack height for rooftop stacks surrounded with architectural screen walls. The SHR is defined as

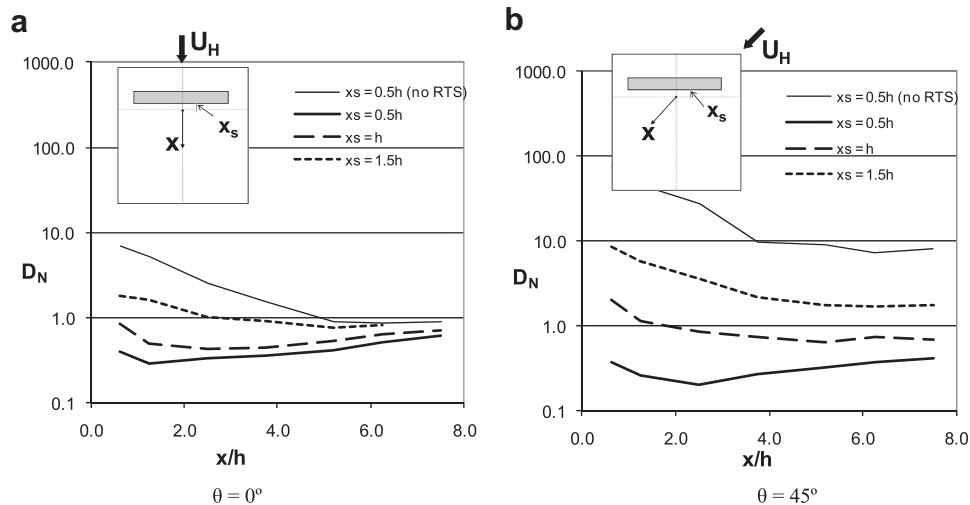


Fig. 14. Effect of separation gap between stack and RTS on along-wind D_N profiles for the low-rise building.

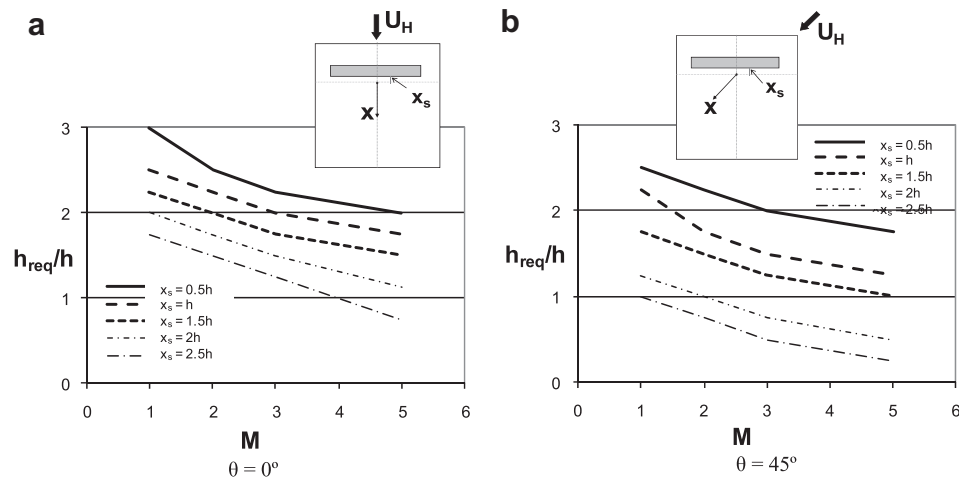


Fig. 15. Stack height required to avoid the downwash effect of an RTS for the low-rise building.

the ratio of stack height without screen divided by stack height with screen.

Although building downwash studies are useful, their findings may not be necessarily applicable to the case of RTS downwash. As shown in this study, increase in RTS width increases rooftop concentrations. However, Huber and Snyder (1989) showed that in the case of building downwash, ground level concentrations decrease with increase in building width, which is a fundamental difference between downwash due to RTS and building.

The present analysis deals with the low-rise building only, as RTS downwash is more significant for this structure than for the high-rise building. A simple design guidance that may be used to determine the stack height required to avoid RTS downwash effect is provided. Fig. 15 shows such data for $\theta = 0^\circ$ and 45° for different M values. By definition, h_{req} is the stack height that gives similar D_N values with and without the RTS. The values of h_{req} were estimated for x_s ranging from $0.5h$ to $2.5h$ with the wide RTS ($w = 7.5h$) that produced maximum downwash effect.

Fig. 15a shows the variation of normalized h_{req} with M for $\theta = 0^\circ$. The values of h_{req} vary significantly with M and x_s . For $x_s = 0.5h$, h_{req} values were $2h$ for $M = 5$ and $3h$ for $M = 1$. As indicated in previous sections, increasing M increases the plume height and thereby reduces plume re-entrainment. Similarly, an increase in x_s decreases plume capture within the RTS downwind cavity. Therefore, h_{req} decreases with an increase in M and x_s .

Fig. 15b shows similar plots for $\theta = 45^\circ$. The trends were similar to the normal wind case. However, h_{req} values are generally smaller than those obtained for $\theta = 0^\circ$. For an oblique wind, the height of the separation bubble formed on the RTS is not significantly affected by an increase in RTS crosswind width. On the other hand, for the normal wind, the height of the separation bubble and length of RTS downwind cavity increases significantly with an increase in RTS crosswind width – see Fig. 4b. Consequently, for $\theta = 45^\circ$, the plume will be discharged outside the RTS recirculation zone for lower values of h_{req} than for $\theta = 0^\circ$.

Charts provided in Fig. 15 may be used as preliminary design guidance for determining stack height to avoid downwash due to an RTS. Results indicate that depending upon RTS-stack separation gap and M ; the required stack height can vary from h to $3h$.

5. Summary and conclusions

Wind tunnel experiments were conducted to quantify the downwash effect of an RTS (RTS) on plume dispersion. Concentration

measurements were obtained on the plume center-line for typical low-rise and high-rise buildings for different stack heights and exhaust speeds with and without a 4 m tall RTS with varying crosswind widths. The results show that the downwash effect of the RTS can significantly alter the flow on the roof of a building. However, the effect is more significant if the RTS is outside the building roof recirculation zone (i.e. it is affected by the upstream flow), which is generally the case with low-rise buildings. The main conclusions from the present study can be summarized as follows:

For low-rise buildings, the downwash effect of an RTS on a plume can significantly decrease dilutions at fresh air intakes located downwind of the stack. The effect increases with RTS crosswind width and for oblique wind directions. The downwash from an RTS decreases the plume height and increases the plume spread. For M values between 1 and 5, an increase in stack height increases roof level dilutions with or without the RTS. The downwash effect decreases significantly with increasing separation distance between stack and RTS. Depending upon RTS-stack separation gap and value of M , the required stack height can vary from h to $3h$ to avoid the downwash effect for an exhaust placed downwind of the RTS.

For the high-rise building, the effect of an RTS on a plume is negligible for the normal wind direction. For M values ranging between 1 and 5, the effect of increasing stack height on roof level dilutions is small irrespective of the RTS presence. For oblique wind directions, the effect is similar to that of low-rise buildings.

The present study focused on the effect of RTS on plume dispersion for isolated buildings. However, the effect of RTS on surrounding buildings or the influence of the surrounding buildings on the effect of RTS on the dispersion of pollutants is also important and will be studied in the future.

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